

François-Pierre Paty

Member of Technical Staff · The Forecasting Company

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Applied mathematician (PhD, École Polytechnique engineer) and Member of Technical Staff at The Forecasting Company, a research lab building foundation models for time-series forecasting. Previously, I built the R&D pipeline at Ida (acquired by RELEX Solutions), published on algorithmic optimal transport at ICML and AISTATS, and taught mathematics at École Centrale de Pékin. I also run a Chinese-learning platform and newsletter reaching more than 25,000 people worldwide.

/ Experience

Sept 2026

– present

Member of Technical Staff

The Forecasting Company · Paris, France

- Research lab building foundation models for time-series forecasting across domains.

Sept 2025

– Aug 2026

Visiting Professor of Mathematics

École Centrale de Pékin (Beihang University) · Beijing, China

- Taught mathematics (analysis, probability, differential calculus and equations) in French to Chinese students: lectures, tutorials and oral examinations (khôlles).
- Presented technical material to varied audiences and collaborated in a multicultural, Chinese-speaking environment.

Oct 2023

– Sept 2025

Research Scientist & R&D Lead (first employee)

Ida (acquired by RELEX Solutions in 2025, now RELEX Fresh) · Paris, France

- Startup optimizing fresh-food orders for grocery retailers. Built the entire R&D pipeline from scratch: research, implementation, production deployment and fast iteration in close contact with field data and data engineers.
- Impact: 30–50% reduction in food waste, 3 margin points gained, deployment across hundreds of stores in Europe.
- Modeling: time-series forecasting (ML), stock reliability (HMM), order optimization (MILP, MDP, dynamic programming).
- Production: deployment and monitoring in Python, Git and CI/CD.
- Mentored MSc-level data-science interns; led technical presentations to clients and in tenders.

May 2022

– present

Founder & creator

HSK Level & Chinese Grammar Gems · Worldwide

- Chinese-proficiency testing platform (120,000+ tests, 5M answers) and a Substack Bestseller newsletter with 25,000+ subscribers worldwide.
- Full-stack development, algorithmic modeling and AI-assisted app building.
- Growing recurring revenue (+50% in 2025); building a new app integrating generative AI.

Sept 2021

– April 2022

Postdoctoral Researcher

CREST, CNRS · Palaiseau, France

- Developed novel machine-learning algorithms with applications in economics; optimal transport applied to labor-market modeling, with numerical simulations in Python (JAX).

Sept 2018

– Aug 2021

PhD Researcher & Teaching Assistant

ENSAE Paris, Institut Polytechnique de Paris · Palaiseau, France

- Algorithmic optimal transport under the supervision of Marco Cuturi (CREST funding).
- 3 publications (ICML ×2, AISTATS) including a Notable Paper Award; 13 talks at international conferences and seminars.
- Teaching assistant in mathematics, optimization, machine learning and deep learning (BSc and MSc levels).

April 2017
– Aug 2017

Research Intern, Applied Mathematics

EDF R&D & Finance for Energy Market Research Centre · Palaiseau, France

- Adapted sparse-deconvolution techniques to missing-data imputation for financial time series. Received the congratulations of the Applied Mathematics department of École polytechnique; a patent was filed by EDF (FR3075969B1).

/ Education

2018

– 2021

PhD in Applied Mathematics and Machine Learning

Institut Polytechnique de Paris (ENSAE) · Palaiseau, France

Thesis: “Optimal transport in high dimension: obtaining regularity and robustness using convexity and projections”, advised by Marco Cuturi.

2017

– 2018

MSc in Statistics and Machine Learning (Mathématiques de l'Aléatoire, M2)

Université Paris-Sud (Paris-Saclay) · Orsay, France

Advanced courses in theoretical statistics and machine learning.

2017

– 2018

Engineering degree (Data Science track)

ENSAE Paris · Palaiseau, France

Specialized in statistics and data science.

2014

– 2018

Ingénieur polytechnicien (X2014)

École Polytechnique · Palaiseau, France

Applied mathematics, with a focus on statistics, probability and data analysis.

2012

– 2014

Classe préparatoire (MP)

Lycée Louis-le-Grand · Paris, France

Intensive mathematics and physics, preparing for the competitive entrance examinations to the Grandes Écoles.

/ Publications

5. Algorithms for Weak Optimal Transport with an Application to Economics

F-P. Paty, P. Choné, F. Kramarz · *Preprint 2022*

4. Optimal transport in high dimension: obtaining regularity and robustness using convexity and projections

F-P. Paty · *PhD thesis, Institut Polytechnique de Paris 2021*

3. Regularized Optimal Transport is Ground Cost Adversarial

F-P. Paty, M. Cuturi · *International Conference on Machine Learning (ICML) 2020*

2. Regularity as Regularization: Smooth and Strongly Convex Brenier Potentials in Optimal Transport

F-P. Paty, A. d'Aspremont, M. Cuturi · *International Conference on Artificial Intelligence and Statistics (AISTATS) 2020* · **Notable Paper Award (top 3 of 423 accepted papers)**

1. Subspace Robust Wasserstein Distances

F-P. Paty, M. Cuturi · *International Conference on Machine Learning (ICML) 2019* · **Oral presentation (top 20% of accepted papers)**

/ Awards & Distinctions

- 2020 · Notable Paper Award (top 3 of 423 accepted papers), International Conference on Artificial Intelligence and Statistics (AISTATS).
- 2019 · Oral presentation (top 20% of accepted papers), International Conference on Machine Learning (ICML).
- 2017 · Congratulations of the Applied Mathematics department, École polytechnique.

/ Talks, Tutorials & Conferences

Conferences

- AISTATS 2020 — contributed talk (online), Aug 2020.
- ICML 2020 — contributed talk (online), Jul 2020.
- ICML 2019 — 20-minute oral presentation, Long Beach, Jun 2019.

Seminars

- New Monge Problems seminar, Université Gustave Eiffel, Jun 2022.
- Mokaplan team seminar, INRIA Paris, May 2022.
- Image, Optimization and Probability seminar, Institut de Mathématiques de Bordeaux, Mar 2021.
- EDMH PhD students seminar, Université Paris-Saclay, Mar 2021.
- Learning meets Astrophysics seminar day, CEA Saclay, Jan 2020.
- Stat-Eco-ML seminar, ENSAE Paris, Nov 2019.
- Le Séminaire Palaisien, INRIA Saclay, Nov 2019.

Summer schools

- Computational optimal transport tutorial, Machine Learning Summer School (MLSS), Moscow, Aug 2019.
- Saint-Flour Probability Summer School, Jul 2019.

Workshops (poster / participant)

- Paris workshop on optimal transport with applications to economics and statistics, Sciences Po Paris, Oct 2021.
- Poster, NeurIPS Optimal Transport and Machine Learning Workshop, Vancouver, Dec 2019.
- People in Optimal Transportation and Applications, Cortona, Jun 2019.
- Poster, Optimization and Statistical Learning workshop, Les Houches, Mar 2019.

/ Teaching

2025

– 2026

École Centrale de Pékin (Beihang University)

- Lectures (cours magistraux, in French) in advanced analysis and probability.
- Tutorials in analysis, differential calculus and differential equations.
- Oral examinations (khôlles).

Dec 2024

École Nationale des Ponts et Chaussées

- Invited course on operations research.

2018

– 2021

ENSAE Paris — Teaching Assistant

- Mathematics (in French): Topology and Functional Analysis (BSc); Differentiable Optimization (BSc); Mathematical Statistics (MSc).
- Computer science (in English): Geometric Methods in Machine Learning (MSc); Stochastic Optimization and Automatic Differentiation for ML (MSc); Optimal Transport (MSc); Deep Learning: Models and Optimization (MSc).

/ Service to the Community

- Conference reviewer: AISTATS 2020, ICML 2020, NeurIPS 2020.
- Co-organizer of the Statistics, Econometrics, Machine Learning (Stat-Eco-ML) seminar, ENSAE Paris, 2019–2021.

/ Skills & Languages

Machine learning

Time series and forecasting, optimal transport, probabilistic models (HMM), deep learning, LLMs and generative AI.

Optimization & OR

MILP, dynamic programming, MDPs, operations research.

Programming

Python (JAX, scikit-learn, PyTorch, cvxpy), SQL, Git, CI/CD, React, Claude Code.

Languages

French (Native) · **English** (Fluent) · **Italian** (Conversational) · **Chinese** (HSK5)